

DPP 03

Computer Science & Information Technology

Discrete Mathematics

Set Theory and Algebra

Q1 Let P be the partial order defined on the set $\{1, 2, 3, 4\}$ as follows

$$P = \{(x, x) \mid x \in \{1, 2, 3, 4\}\} \cup \{(1, 2), (3, 2), (3, 4)\}$$

The number of total orders on $\{1, 2, 3, 4\}$ that contains P is _____.

Q2 Which of the following is/are always true for any lattice?

- (A) There exists exactly one minimum and exactly one maximum element.
- (B) There exists at most one minimal and at most one maximal element.
- (C) Least upper bound and greatest lower bound exists for every pair of elements.
- (D) Every element has a unique complement.

Q3 Let $f: A \rightarrow B$ be a function, and let E and F be subsets of A . Consider the following statements.

$$S1: f(E \cup F) = f(E) \cup f(F)$$

$$S2: f(E \cap F) \subseteq f(E) \cap f(F)$$

$$S3: f(E \cap F) = f(E) \cap f(F) \text{ if function } f \text{ is one-one.}$$

Which of the following is true?

- (A) Only $S2$ and $S3$ are true.
- (B) Only $S1$ and $S3$ are true.
- (C) Only $S1$ and $S2$ are true.
- (D) All $S1$, $S2$ and $S3$ are true.

Q4 Let $f(x) = \frac{x}{x+1}$ and $g(x) = \frac{x}{1-x}$ then $(f \circ g)^{-1}(x) = ?$

- (A) x
- (B) $x-1$
- (C) $x+1$
- (D) $x+2$

Q5 Let $A = \mathbb{R} - \{3\}$

$$B = \mathbb{R} - \{1\}$$

Where $\mathbb{R}$ is the set of all real numbers if

$f: A \rightarrow B$ such that $f(x) = \frac{x-2}{x-3}$, then

(A) f is one-one but not onto

(B) f is not one-one but onto

(C) f is bijection

(D) f is neither one-one nor onto

Q6 If the function $f: [1, \infty) \rightarrow [1, \infty)$ defined by $f(x) = 2^{x(x-1)}$ is invertible, then $f^{-1}(x)$ is :

- (A) $\frac{1 + \sqrt{1 + 2 \log_2 x}}{4}$
- (B) $\frac{1 - \sqrt{1 + 2 \log_2 x}}{4}$
- (C) $\frac{1 + \sqrt{1 + 4 \log_2 x}}{2}$
- (D) $\frac{1 - \sqrt{1 + 4 \log_2 x}}{2}$

Q7 The domain of the function:

$$f(x) = \sin \left\{ \log \left(\frac{\sqrt{4-x^2}}{(1-x)} \right) \right\} \text{ is :}$$

- (A) $(-2, 0)$
- (B) $(-2, 1)$
- (C) $(-2, 2) - \{1\}$
- (D) $(0, 2) - \{1\}$

Q8 Which of the following statements is /are True?

- (A) A constant function is one-one iff domain of the functions has exactly one element
- (B) A constant function is onto iff co-domain of the functions has exactly one element
- (C) Every one-one function on a finite set A is bijection
- (D) Every onto function on a finite set A is bijection

Q9 Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$

Set C is defined as $C = \{f: A \rightarrow B \mid 2 \in f(x) \text{ for some } x \in A, \text{ and } f \text{ is not one-one}\}$

Then number of elements in set C are _____.

Q10 Let $|A| = 6$ and $|B| = 3$, then number of onto functions possible from set A to set B is



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Answer Key

Q1 5~5

Q2 (B, C)

Q3 (D)

Q4 (A)

Q5 (C)

Q6 (C)

Q7 (B)

Q8 (A, B, C, D)

Q9 19~19

Q10 540~540

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Hints & Solutions

Q1 Text Solution:

The number of total orders on $\{1, 2, 3, 4\}$ that contains P is 5.

Q2 Text Solution:

There exists at most one minimal and at most one maximal element.

Least upper bound and greatest lower bound exists for every pair of elements.

Q3 Text Solution:

All S1, S2 and S3 are true.

Q4 Text Solution:

$$f\left(\frac{x}{x+1}\right), g\left(\frac{x}{1-x}\right) = \frac{x}{1-x}$$

$$f \circ g\left(\frac{x}{1-x}\right) = f\left(\frac{\frac{x}{1-x}}{1+\frac{x}{1-x}}\right) = \frac{\frac{x}{1-x}}{\frac{1-x+x}{1-x}} = \frac{x}{1-x} = x$$

Thus, $f(g(x)) = x$ which means fog is the identity function.

$$\text{So, } (f \circ g)^{-1}(x) = x$$

Q5 Text Solution:

$$\text{Let } f(x_1) = f(x_2)$$

$$\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3}$$

$$x_1 x_2 - 3x_1 - 2x_2 + 6 = x_1 x_2 - 2x_1 - 3x_2 + 6$$

$$x_1 = x_2$$

Hence, f is one – one.

$$f(x) : y = \frac{x-2}{x-3}$$

$$xy - 3y = x - 2$$

$$xy - x = 3y - 2$$

$$x = \frac{3y-2}{y-1}$$

$$x \in \mathbb{R} - \{1\}$$

Since $y \in B = \mathbb{R} - \{1\}$, x is well-defined for all $y \in B$.

Thus, f is onto.

Since f is both one-one and onto, it is a bijection.

Q6 Text Solution:

$$y = 2^{x(x-1)}$$

$$\log_2 y = x(x-1)$$

$$x^2 - x - \log_2 y = 0$$

$$x = 1 \pm \frac{\sqrt{1+4\log_2 y}}{2}$$

In terms of x it can be written as $\frac{1 \pm \sqrt{1+4\log_2 x}}{2}$

As the range is $\left[1, \infty\right)$, $\frac{1 - \sqrt{1+4\log_2 x}}{2}$ will be

invalid.

Hence, option c is correct.

Q7 Text Solution:

$$f\left(\frac{x}{1-x}\right) = \sin\left\{\log\left(\frac{\sqrt{4-x^2}}{(1-x)}\right)\right\}$$

Due to log, $\frac{\sqrt{4-x^2}}{(1-x)} > 0$ must hold true.

Also because of square root,

$$4 - x^2 > 0 \text{ must hold true}$$

$$= (2+x)(2-x) > 0$$

$$= -2 < x < 2$$

As numerator is positive, hence the denominator must also be positive.

$$\text{Thus, } 1 - x > 0$$

$$\Rightarrow x < 1$$

Hence, domain of the function is $(-2, 1)$.

Q8 Text Solution:

(A) A constant function maps every element in the domain to the same single element in the co-domain. This means it cannot be one-one if the domain has more

than one element, because multiple elements in the domain would map to the same element in the co-domain.

(B) A constant function maps every element in the domain to a single element in the co-domain. For the function to be onto, the co-domain must contain only that

single element; otherwise, it cannot cover all elements in the co-domain.

(C) A one-one function (injective function) on a finite set A maps each element of A uniquely to the co-domain. If the function is also a bijection, it must be both

one-one and onto, meaning the co-domain must be exactly the range of A.

(D) An onto function (surjective function) on a finite set A covers the entire co-domain. Since the function is onto and A is finite, it must also be one-one (injective)



because otherwise, there would be elements in the co-domain not covered

Q9 Text Solution:

Total number of functions from $A \rightarrow B = 4^3 = 64$

Number of injection functions from $A \rightarrow B = 4 \times 3 \times 2 = 24$

Thus, the number of non-injective function $= 64 - 24 = 40$

If 2 is not in the range, each element in A can map only to {1, 3, 4}.

The number of non-one-one functions mapping A to {1, 3, 4}:

- Total functions from A to {1, 3, 4} is $3^3 = 27$

- One-one functions from A to {1, 3, 4} is $3 \times 2 \times 1 = 6$

- So, non-one-one functions mapping A to {1, 3, 4} is $27 - 6 = 21$.

Number of Non-one-one functions with $2 \in \text{range}(f)$
 $= 40 - 21 = 19$.

Q10 Text Solution:

Number of onto functions = Total functions –

Exclusion of missing one + Inclusion of missing two

$$= 3^6 - 3 \cdot 2^6 + 3 \cdot 1^6$$

$$= 729 - 192 + 3$$

$$= 540$$


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